12th HACKERS KA JUGAAD

Electric charges & Fields

Electric charge — charge is the property associated with matter due to which it produces and experiences electrical and magnetic effects. The excess or deficiency of electrons in a body gives the concept of charge.

atoms → intrinsic → electric and magnetic effect.



SI unit of charge: ampere second i.e. Coulomb

CGS unit of charge: electrostatic unit of charge (e.s.u. of charge) or statcoulomb (stat C).

$1C = 3 \times 10^9$ stat coulomb $= 3 \times 10^9$ e.s.u. of charge

Electromagnetic CGS unit of charge: ab coulomb or electromagnetic unit of charge (e.m.u. of charge).

1 coulomb $= \frac{1}{10}$ ab coulomb $= \frac{1}{10}$ e.m.u. of charge

Practical units of charge are ampere hour (= 3600C) and faraday (= 96500C)

$I = \frac{q}{t} \Rightarrow q = I \times t$ (A, h)

- ✓ Charge is a scalar quantity.
- ✓ Charge is transferable.
- ✓ Charge is always associated with mass.
- ✓ Charge is quantised.
- ✓ Charge is conserved.
- ✓ Charge is invariant.
- ✓ Accelerated charge radiates energy.
- ✓ Similar charges repel each other while dissimilar attract.

Proton → +ve
$q_p = 1.6 \times 10^{-19} C$
$m_p = 1.67 \times 10^{-27} kg$

Electron → -ve
$q_e = -1.6 \times 10^{-19} C$
$m_e = 9.1 \times 10^{-31} kg$

Neutron → 0
$q_n = 0C$
$m_n = 1.67 \times 10^{-27} kg$

Quantization of charge – the quantization of electric charge means that the total charge (q) of a body is always an integral multiple of a basic quantum of charge (e) i.e.

$$q = ne \quad \text{where} \quad n = 0, \pm 1, \pm 2, \pm 3, \ldots$$

for $n = 1$

min. amount of charge that can be transferred

$$q = e$$

$$1\,C = ne$$

$$q = ne$$

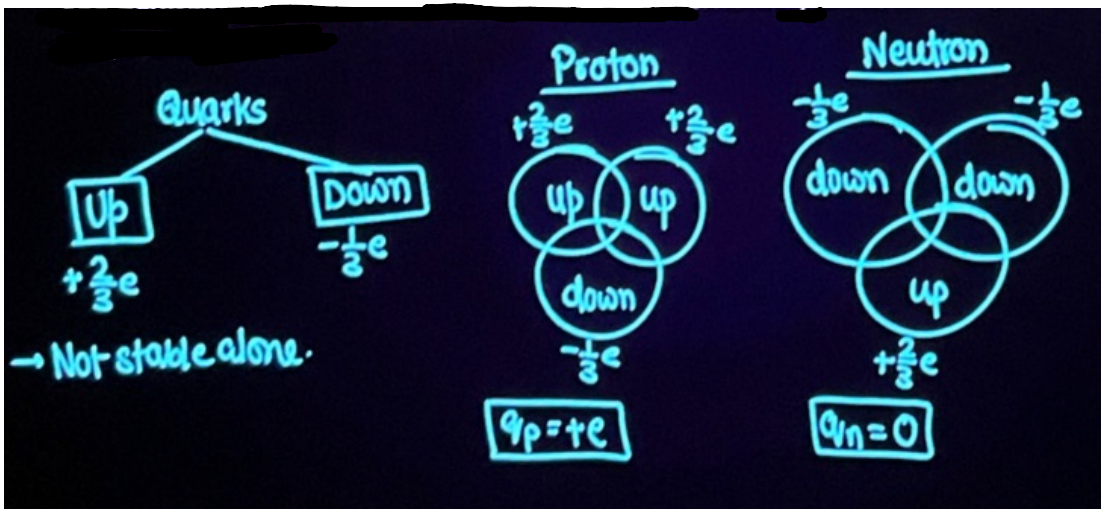
$$1 = n(1.6 \times 10^{-19})$$

$$n = \frac{1}{1.6 \times 10^{-19}} = \frac{1 \times 10^{19}}{1.6} = \frac{10}{1.6} \times 10^{18}$$

$$= 6.25 \times 10^{18}.$$

Quarks –

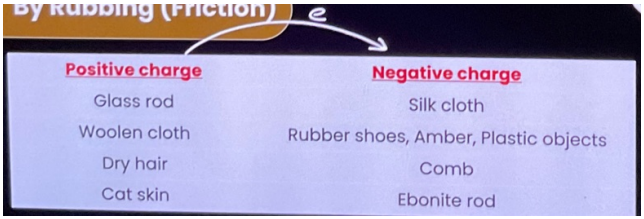
Recent discoveries in high-energy Physics have indicated that the elementary particles like protons and neutrons are themselves built out of more elementary units called quarks which have charges $2/3\,e$ and $-1/3\,e$



Methods of charging –

By Rubbing (friction)

If we rub one body with other body electrons are transferred from one body to the other transfer of electron takes place from lower work function body to higher work function body



By touching (conduction)

The process of transfer of charge by contact of two bodies is known as conduction. If a charge body is put in contact with uncharged body the uncharged body becomes charged due to transfer of electrons from one body to the other.

The charged body loses some of its charge which is equal to the charge gained by the uncharged body.

The charge gained by the uncharged body is always lesser than initial charge present on the charged body.

Flow of charge depends upon the potential difference of both bodies no potential difference means No conduction

Positive charge flows from higher potential to lower potential, while negative charge flows from lower to higher potential

By Induction –

If a charged body is brought near a metallic neutral body, the charged body will attract opposite charge and ripple similar charge present in the neutral body. As a result of this one side of the neutral body becomes negative while the other positive this process is called electrostatic induction.



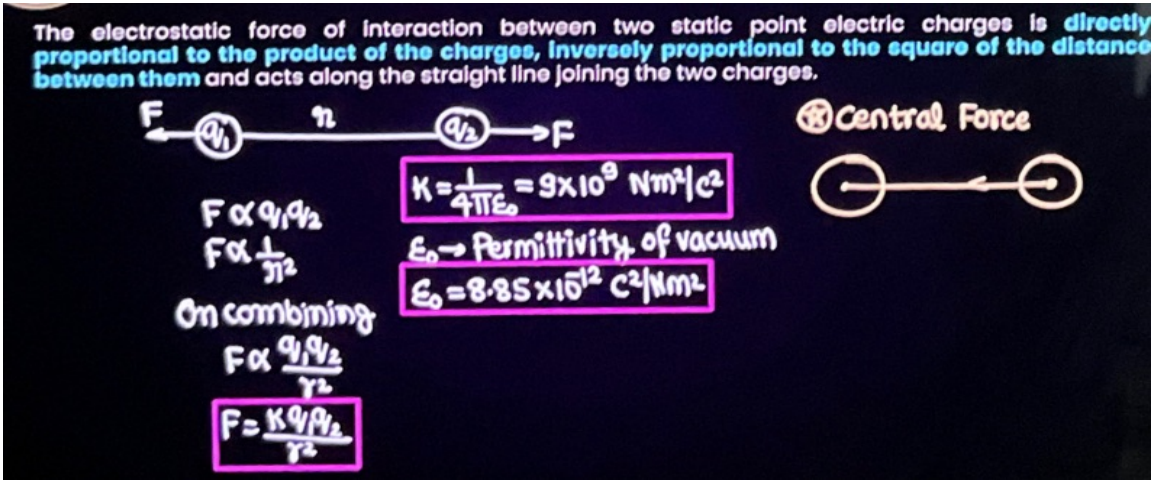
Some of the important facts associated with induction

1.Reducing body, neither gains nor loses charge

2.The nature of induced to charge is always opposite to that of inducing charge

3.Induction takes place only in conducting bodies and not in particles

Coulomb's law –



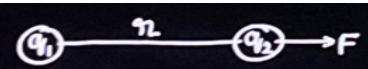
In medium,

$\varepsilon = K\varepsilon_0$

Permittivity of medium — Dielectric Constant (Insulator)

$K = \frac{\varepsilon}{\varepsilon_0}$

ε_0 (Relative Permittivity)



In vacuum,

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$$

$K_{air} \approx 1$

$K_{vacuum} = 1$

$K_{water} = 80$

In medium,

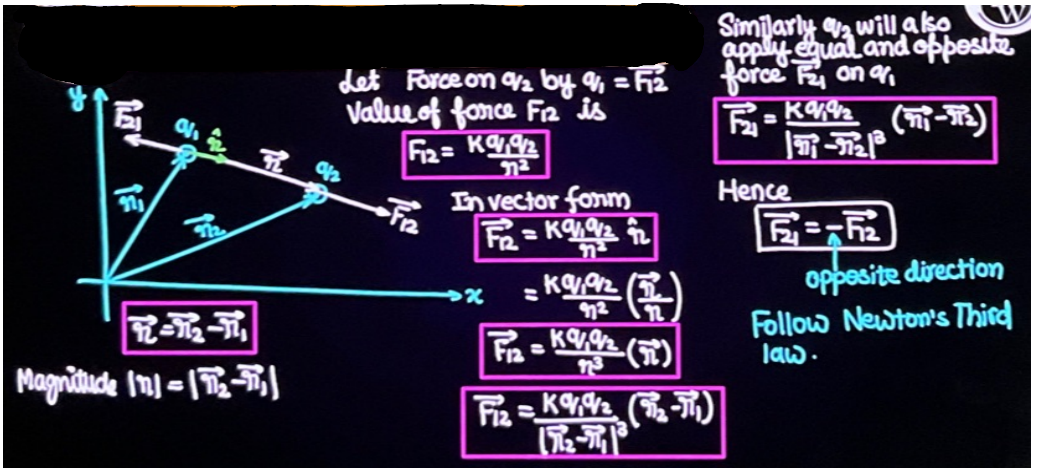
$$F' = \frac{1}{4\pi\varepsilon}\frac{q_1 q_2}{r^2}$$

$$F' = \frac{1}{4\pi(\varepsilon_0 K)}\frac{q_1 q_2}{r^2}$$

$$F' = \frac{F}{K}$$

Dielectric Constant

Coulomb's law in vector form

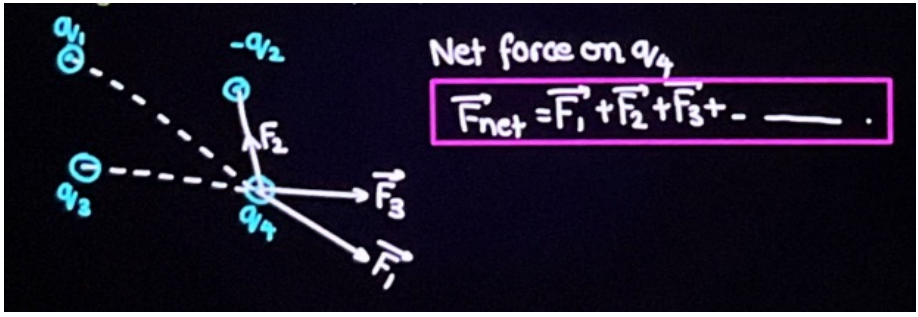


Limitation of Coulomb's law-

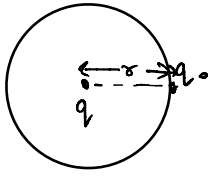
1. The electric charges must be at rest.
2. The electric charges must be point charges. The extension of charges must be much smaller than the separation between the charges.
3. The separation between the charges must be greater than the nuclear size because for distances $< 10^{-15}$ m the strong nuclear force dominates over the electrostatic force.

The superposition principle

The principle of superposition states that when a number of charges are interacting, the total force on a given charge is the vector sum of the forces exerted on it due to all other charges. The force between two charges is not affected by the presence of other charges.



Electric field



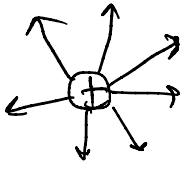
$E = \frac{F}{q_0}$ unit = N/C

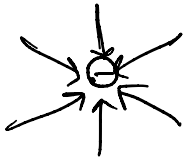
vector quantity

q_0 → test charge (v.v small)

(It should have negligible EF)

$E = \frac{Kq_1 q_0}{r^2} / q_0 = \frac{Kq}{r^2}$





In $\vec{E}$

IC $E = \frac{Kq}{r^2} = 9 \times 10^9$ N/C

⊕ $q_0 \rightarrow \vec{E}$ away from +ve charge

⊖ $\vec{E} \leftarrow q_0$ towards -ve charge

Electric field or the electric intensity or the electric field strength at a point is defined as the force experienced by a unit positive test charge placed at that point without disturbing the position of source charge

The electric field at a point is defined as the electrostatic force per unit test charge acting on a vanishingly small positive test charge placed at that point

$$\vec{E} = \frac{\vec{F}}{q_0}$$

Electric field due to a point charge

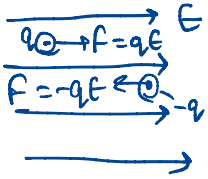


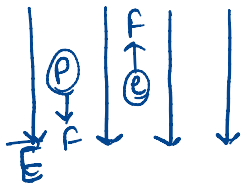
→ vector
→ N/C

$$F = \frac{kq_1 q_0}{r^2} \qquad E = \frac{F}{q_0} = \frac{kq}{r^2}$$



Concept of force in electric field.





Concept of Motion -

- uniform
  - const. speed
  - $a = 0$
  - $v = d/t$
- non-uniform
  - variable speed
  - accelerated motion
  - $a = dv/dt$

a = constant

eq^ of motion

$v = u + at$

$s = ut + \frac{1}{2}at^2$

$v^2 = u^2 + 2as$

a = variable.

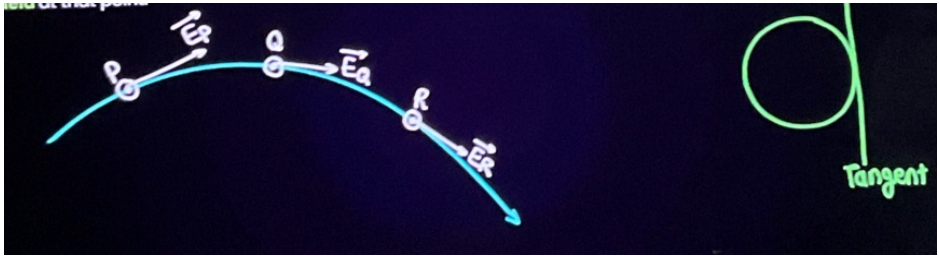
$v = \frac{ds}{dt}$

$a = \frac{dv}{dt}$

$a = v.\frac{dv}{ds}$.

Electric field lines

And electric line of force may be defined as the curve along which a small positive charge would tend to move when free to do so in an electric field and the tangent to which at any point gives the direction of the electric field at that point.



Properties of electric field lines

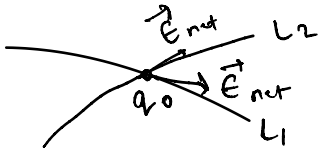
1. The lines of force or continuous, smooth curves without any breaks.

2. The lines of force start at positive charges and end at negativecharges. They cannot form close loops. If there is a single charge, then the lines of force will start or end at infinity.



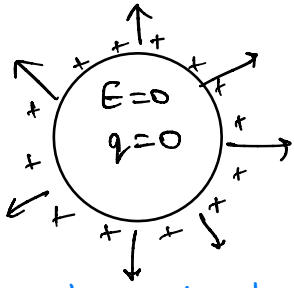
3.The tangent to a line of force at any point gives the direction of the electric field at that point

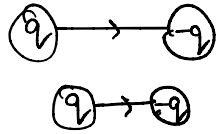
4.No, two lines of force can cross each other



5.The lines of force are always normal to the surface of a conductor on which the charges are in equilibrium

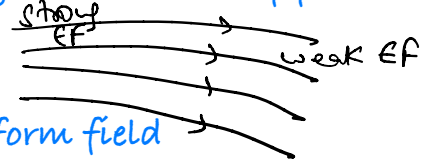
6.The lines of force have a tendency to contract length wise. This explains attraction between two unlike charges.



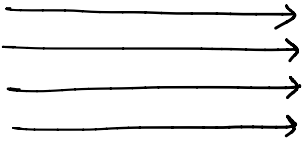


7.The lines of force have a tendency to expand laterally so as to exert a lateral pressure on neighbouring lines of force. This explains repulsion between two similar charges

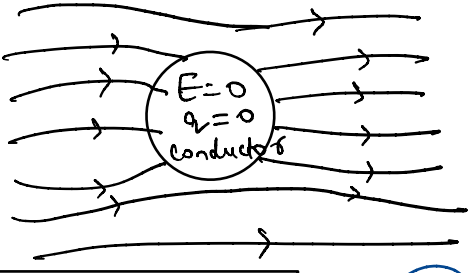
8.The relative closeness of the lines of force gives a measure of the strength of the electric field in any region. The lines of force are
-Close together in a strong field
-Far apart in a weak field
-Parallel and equally spaced in a uniform field



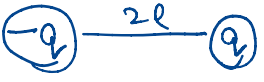
9.The lines of force do not pass through a conductor because the electric field inside a charged conductor is zero



uniform EF
↓
constant



Electric Dipole



$p = q(2\ell)$

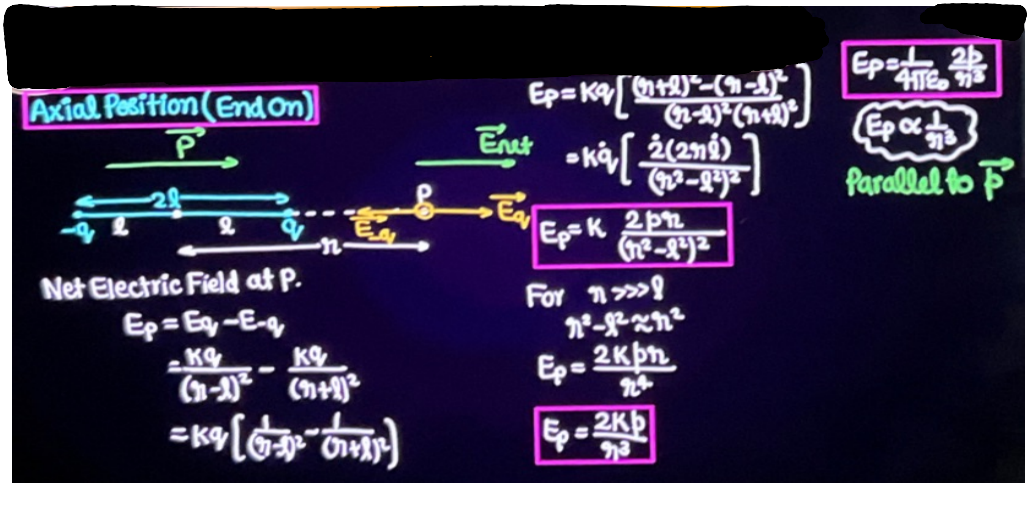
Dipole moment

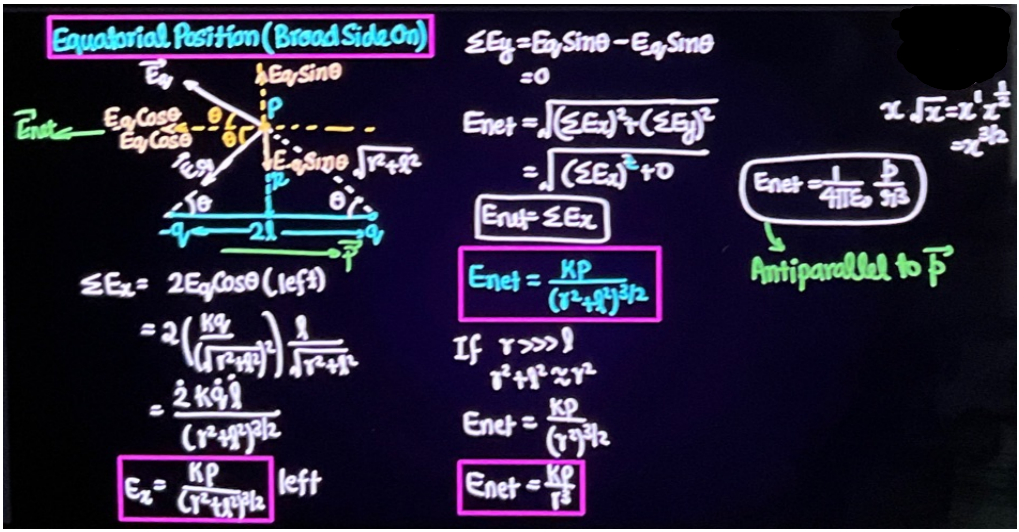
$\vec{p} = q(\vec{2\ell})$

vector quantity

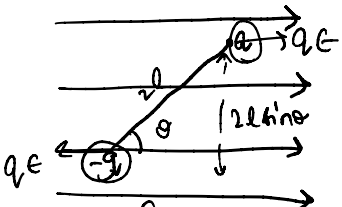
-ve → +ve

Electric field due to Electric dipole





Torque on Dipole in Electric field



torque or couple is created by equal and opposite forces on any body.

τ = F x ⊥ distance

$qE \times 2l\sin\theta$

$\tau = pE\sin\theta$

$\theta = 0°$ or $180°$

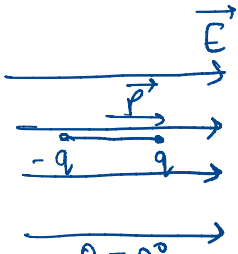
$\tau = 0$

$\theta = 90°$

$\tau_{max} = pE$

In vector form,

$\vec{\tau} = \vec{p} \times \vec{E}$

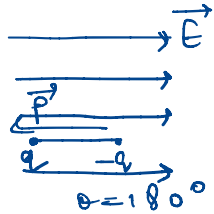


$\tau = pE\sin\theta = 0$

$F_{net} = 0$

stable equilibrium

↳ U_{min}



$\tau = pE\sin\theta = 0$

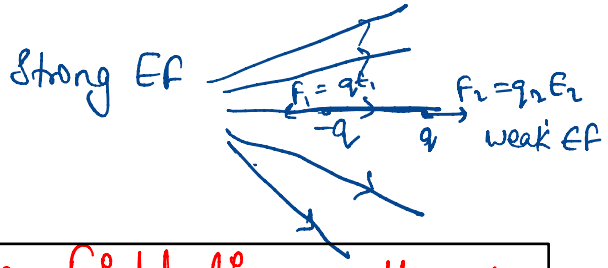
$F_{net} = 0$

Unstable equil.

↓

U_{max}.

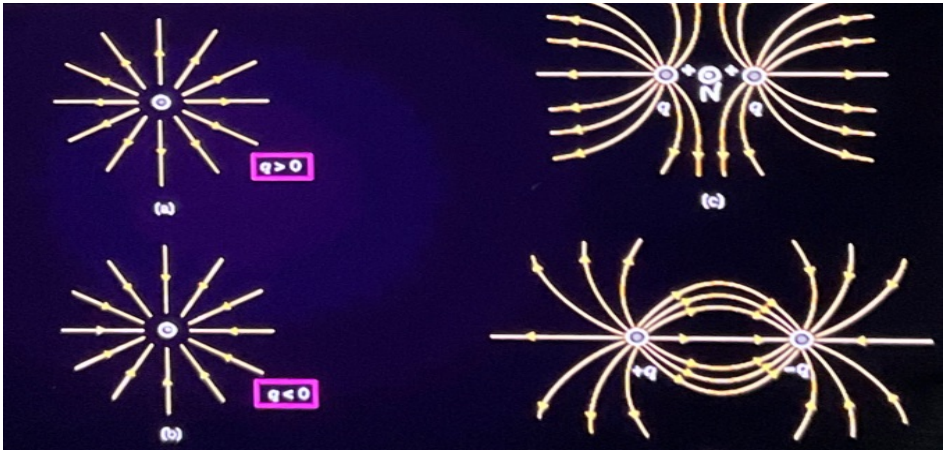
non uniform electric field -



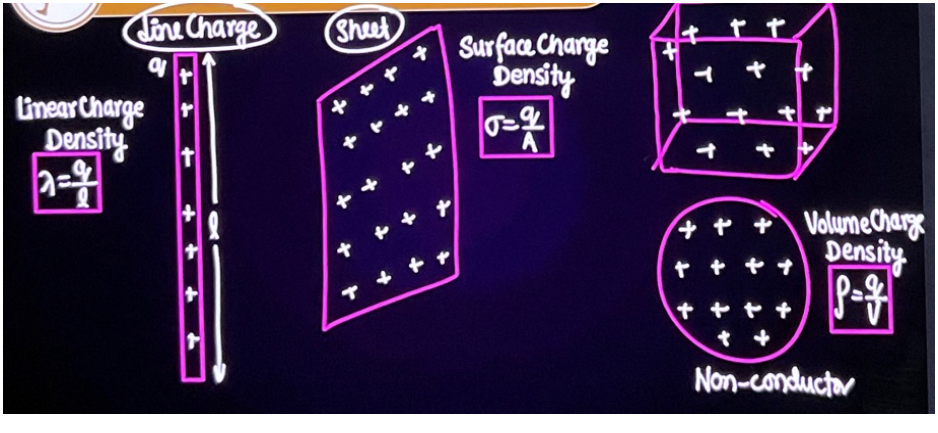
$F_{net} \neq 0$

$\tau \neq 0$

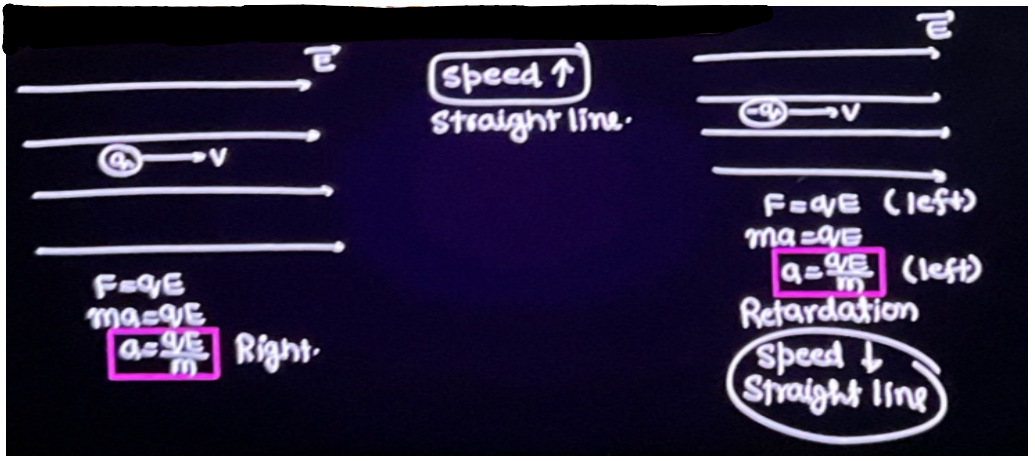
Electric field line patterns

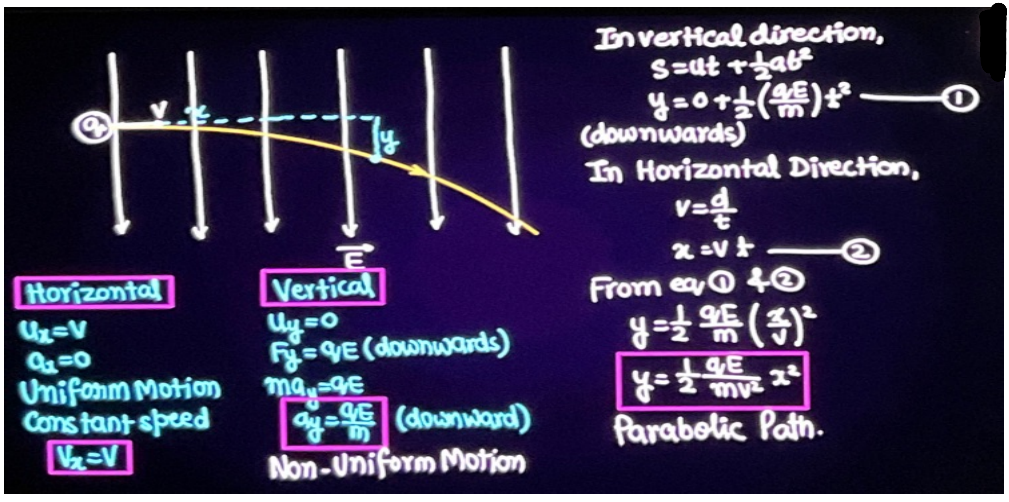


Continuous charge distributions

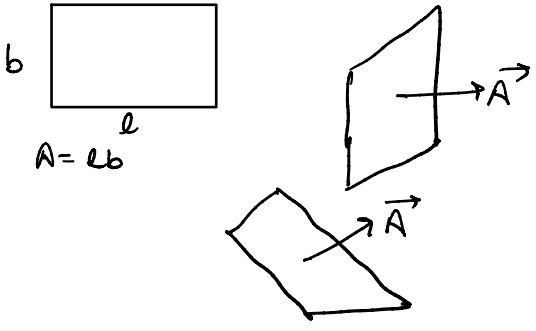


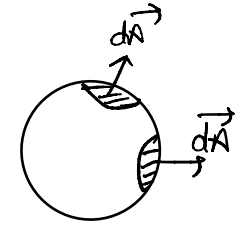
Motion of charge in Electric field





Area Vector

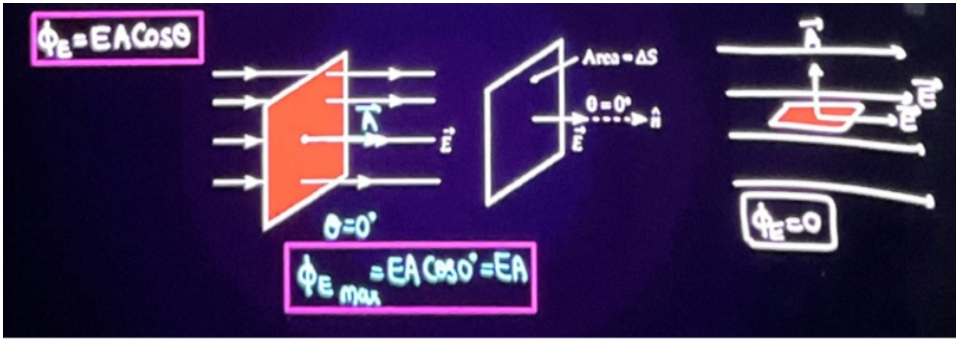


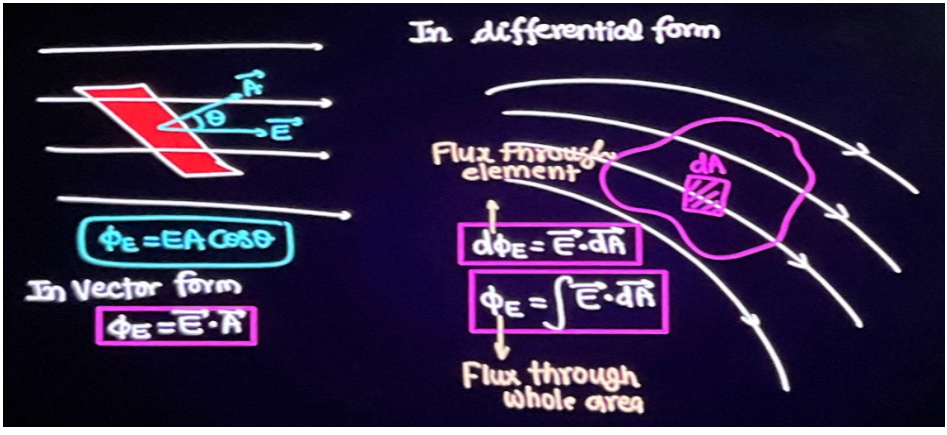


$\vec{A} = A\,\hat{n}$ perpendicularly outwards

Electric flux

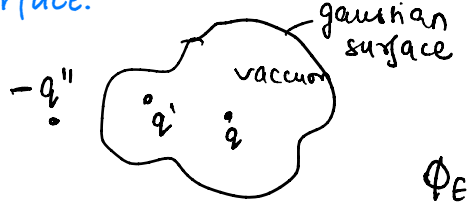
The electric flux through a given area head inside an electric field is the measure of the total number of electric lines of force passing normally through that area.





Gauss Law

This theorem gives a relationship between the total flux passing through any closed surface and the net charge enclosed within the surface. Gauss theorem states that the total flux through a closed surface is $1/\epsilon_0$ times the net charge enclosed by the closed surface.



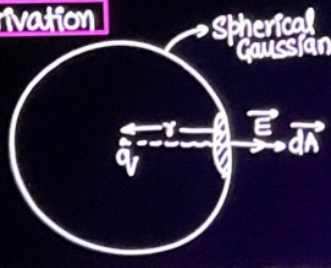
$$\phi_E = \oint_s \vec{E}.\vec{ds} = \frac{q}{\epsilon_0}$$

$$\phi_E = \frac{q+q'}{\epsilon_0}$$

In any medium → $\phi_E = \frac{q+q'}{K\varepsilon_0}$

└ dielectric constant.

Derivation



Flux passing through element

$d\phi_E = \vec{E} \cdot \vec{dA}$

$d\phi_E = E\,dA\cos 0°$

$d\phi_E = E\,dA$

Now for complete gaussian surface,

$\phi_E = \int d\phi_E$

$= \int E\,dA$

$= \int \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\,dA$

$= \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\int dA$

$= \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}(4\pi r^2)$

$\phi_E = \frac{q}{\varepsilon_0}$

GAUSS LAW

Electric field at any point on S is

$$E = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q}{r^2}$$

This field points radially outward at all points on S. Also, any area element points radially outwards, so it is parallel to $\vec{E}$, i.e., $\theta = 0°$.

∴ Flux through area $\vec{dS}$ is

$$d\phi_E = \vec{E}\cdot\vec{dS} = E\,dS\cos 0° = E\,dS$$

Total flux through surface S is

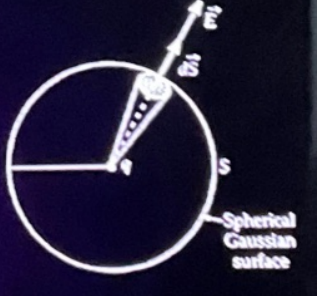
$$\phi_E = \oint_S d\phi_E = \oint_S E\,dS = E\oint_S dS$$

$\phi_E = E \times$ Total area of sphere

$$\phi_E = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q}{r^2}\cdot 4\pi r^2$$

$$\phi_E = \frac{q}{\varepsilon_0}$$

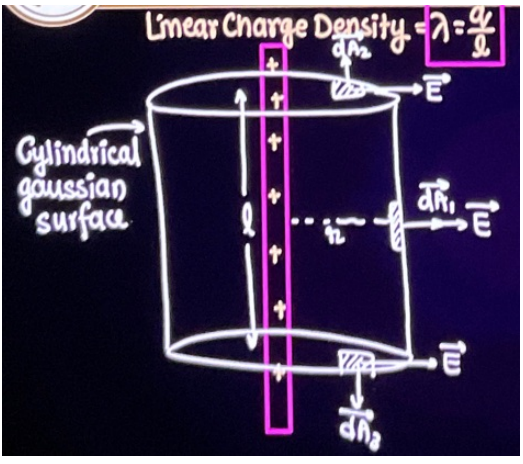
This proves Gauss's theorem.



Electric field due to line charge

net electric flux through gaussian surface,

$$\phi_E = \int \vec{E}\cdot\vec{dA} = \frac{q_{in}}{\varepsilon_0}$$



$$\int \vec{E}.d\vec{A}_1 + \int \vec{E}.d\vec{A}_2 + \int \vec{E}.d\vec{A}_3 = \frac{q_{in}}{\epsilon_0}$$

$$\int E\,dA.\cos\theta + \int E dA_2 \cos 90^\circ + \int E dA_3 \cos 90^\circ = \frac{\lambda l}{\epsilon_0}$$

$$E\int dA_1 + 0 + 0 = \frac{\lambda l}{\epsilon_0}$$

$$E(2\pi r l) = \frac{\lambda l}{\epsilon_0}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \Rightarrow E \propto \frac{1}{r}$$

Electric field due to charged sheet.

By Gauss law,

$$\Phi_E = \frac{q_{in}}{\epsilon_0}$$

$$\int \vec{E}.d\vec{A}_1 + \int \vec{E}.d\vec{A}_2 + \int \vec{E}.d\vec{A}_3 + \int \vec{E}.dA_4 = \frac{q_{in}}{\epsilon_0}$$

$$E\int dA_1 + 0 + E\int dA_3 + 0 = \frac{q_{in}}{\epsilon_0}$$

$$EA + EA = \frac{\sigma A}{\epsilon_0}$$

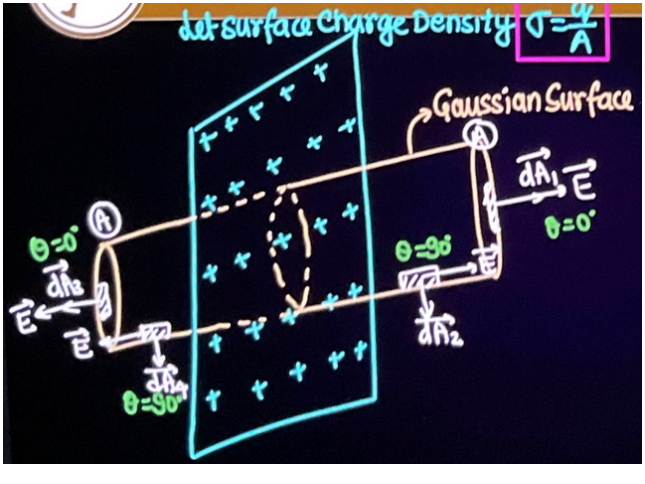
$$2E\cancel{A} = \frac{\sigma \cancel{A}}{\epsilon_0}$$

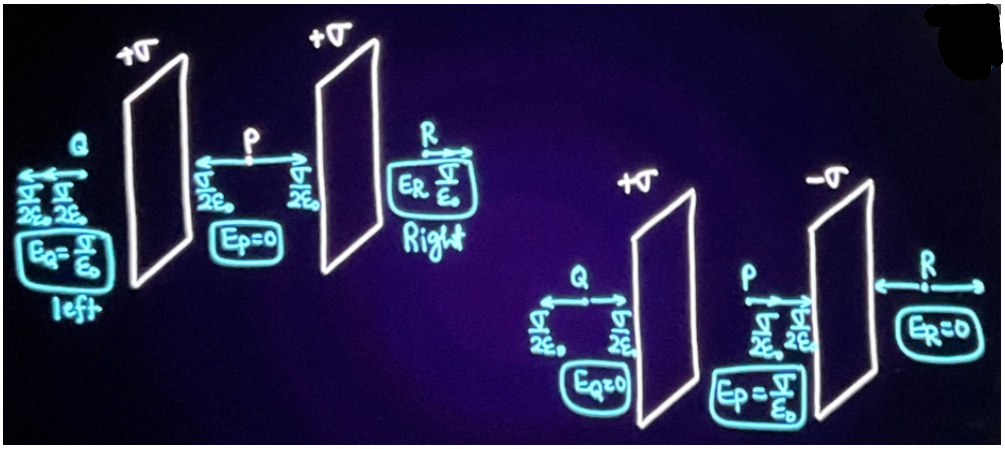
$$\boxed{E = \frac{\sigma}{2\epsilon_0}}$$

Independent of distance & size of sheet

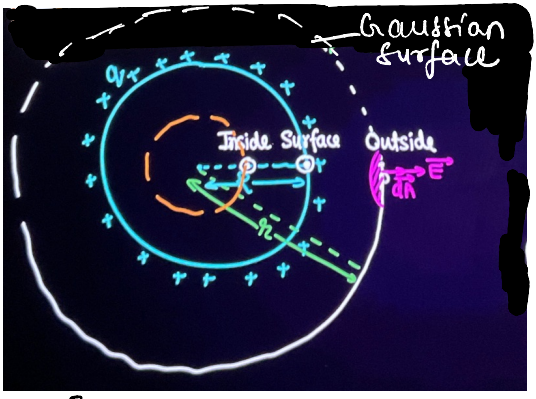
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away from sheet.





Electric Field due to thin spherical shell



for outside point.

By gauss law,

$$\phi_E = \frac{q_{in}}{\varepsilon_0}$$

$$\int \vec{E}\cdot d\vec{A} = \frac{q}{\varepsilon_0}$$

$$E\int dA = q/\varepsilon_0$$

$$E(4\pi r^2) = q/\varepsilon_0$$

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} \qquad E_0 \propto \frac{1}{r^2}$$

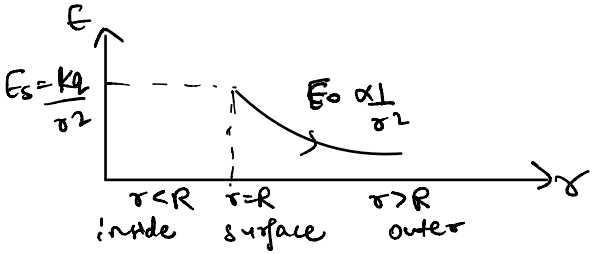
on surface $r = R$

$$E_s = \frac{1}{4\pi\varepsilon_0}\frac{q}{R^2}$$

for inside point

$$\int \vec{E}\cdot d\vec{A} = \frac{q_{in}}{\varepsilon_0}$$

$$E_{in} = 0.$$



THANK YOU !!